Question	Answers	Extra information	Mark	AO / Spec Ref.
02.6	more females were vaccinated (over time)	allow males may also be vaccinated allow more people were vaccinated allow increased use of (named) barrier methods of contraception allow more awareness / education (about HPV)	1	AO3 4.2.2.7 4.2.2.5

Question	Answers	Extra information	Mark	AO / Spec Ref.
02.7	white blood cells		1	AO1 4.3.1.7

Question	Answers	Extra information	Mark	AO / Spec Ref.
02.8	antibodies		1	AO1 4.3.1.7 4.3.1.6

Question	Answers	Extra information	Mark	AO / Spec Ref.
02.9	any **one** from: • people are afraid of side / unknown effects • religious / cultural objections • (some people) believe they don't work • some people think (HPV) vaccine encourages sexual activity	ignore pain of injection allow there are side effects allow people think they cause (named) disease ignore religion unqualified ignore moral / ethical objections	1	AO3 4.3.1.7

Total Question 2	**11**

Question 3

Question	Answers	Extra information	Mark	AO / Spec Ref.
03.1	*(lhs)* carbon dioxide water *(rhs)* glucose	allow word(s) take precedence over formulae } in either order	 1 1 1	AO1 4.4.1.1

Question	Answers	Extra information	Mark	AO / Spec Ref.
03.2	(used in aerobic) respiration to release / transfer energy	do **not** accept anaerobic allow for energy allow (respiration) by mitochondria do **not** accept energy created / made / produced	1 1	AO1 4.4.1.2 4.4.2.1

Question	Answers	Extra information	Mark	AO / Spec Ref.
03.3	any **one** pair from: • bulb / lamp is hot (so) may burn you / skin • glassware is breakable (1) (so) may cut skin (1) • electricity in close proximity to water (1) (so) may get electric shock (1) • scissors / scalpel (to cut pondweed) are sharp (1) may cut skin (1)	mark in pairs allow named example of electrical item in close proximity to water	1 1	AO2 4.4.1.2 RPA6

Question	Answers	Extra information	Mark	AO / Spec Ref.
03.4	(use a) water bath	allow put beaker into (another) beaker of water allow other correct methods eg heat shield allow use an LED	1	AO2 4.4.1.2 RPA6

Question	Answers	Extra information	Mark	AO / Spec Ref.
03.5	(change in) temperature affects / changes (the rate of) photosynthesis	allow temperature is a limiting factor (of photosynthesis) allow temperature affects enzyme (activity) allow temperature affects rate of reaction	1	AO2
	(so) will affect / change the number of bubbles (produced)	allow (so) will affect results allow (so) will affect validity ignore to make it fair ignore so the results will be the same	1	AO3 4.4.1.2 RPA6

Question	Answers	Extra information	Mark	AO / Spec Ref.
03.6	blue		1	AO3 4.4.1.2 RPA6

Question	Answers	Extra information	Mark	AO / Spec Ref.
03.7	bar graph		1	AO3 4.4.1.2 RPA6

Question	Answers	Extra information	Mark	AO / Spec Ref.
03.8	any **one** from: • measuring cylinder • (gas) syringe • burette		1	AO3 4.4.1.2 RPA6